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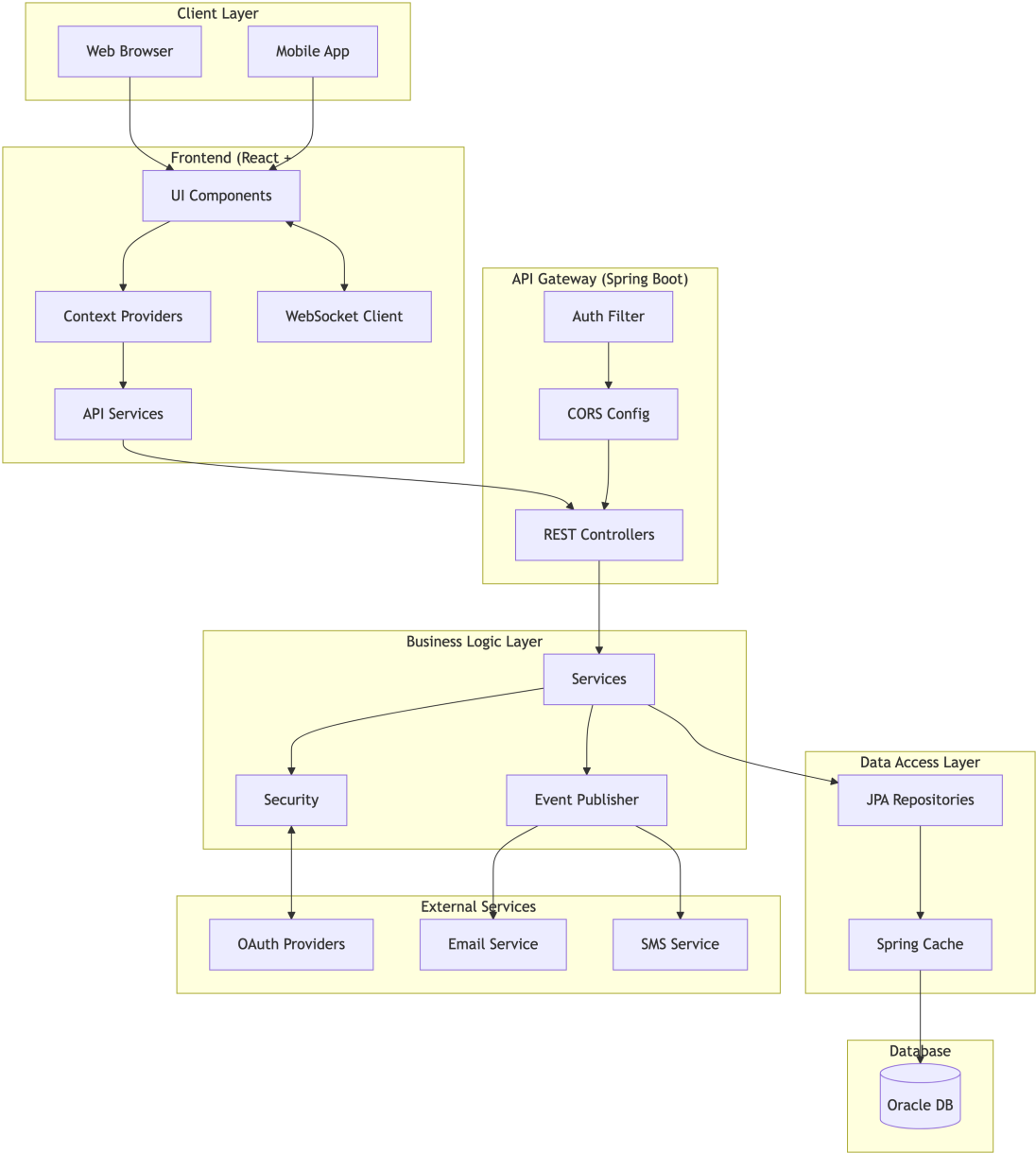
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System Architecture - Gym Management System

High-Level Architecture



Technology Stack

Frontend

Technology	Version	Purpose
React	18.x	UI Library
TypeScript	5.x	Type Safety
Vite	5.x	Build Tool
React Router	6.x	Client Routing
Axios	1.x	HTTP Client
Socket.io Client	4.x	Real-time Communication
Framer Motion	10.x	Animations
Lucide React	-	Icons
Recharts	2.x	Data Visualization

Backend

Technology	Version	Purpose
Spring Boot	3.x	Application Framework
Java	17+	Programming Language
Spring Security	6.x	Authentication & Authorization
Spring Data JPA	3.x	Data Access
Spring WebSocket	-	Real-time Messaging
Hibernate	6.x	ORM
Lombok	1.18.x	Boilerplate Reduction
JWT	0.11.x	JWT Handling

Database & Infrastructure

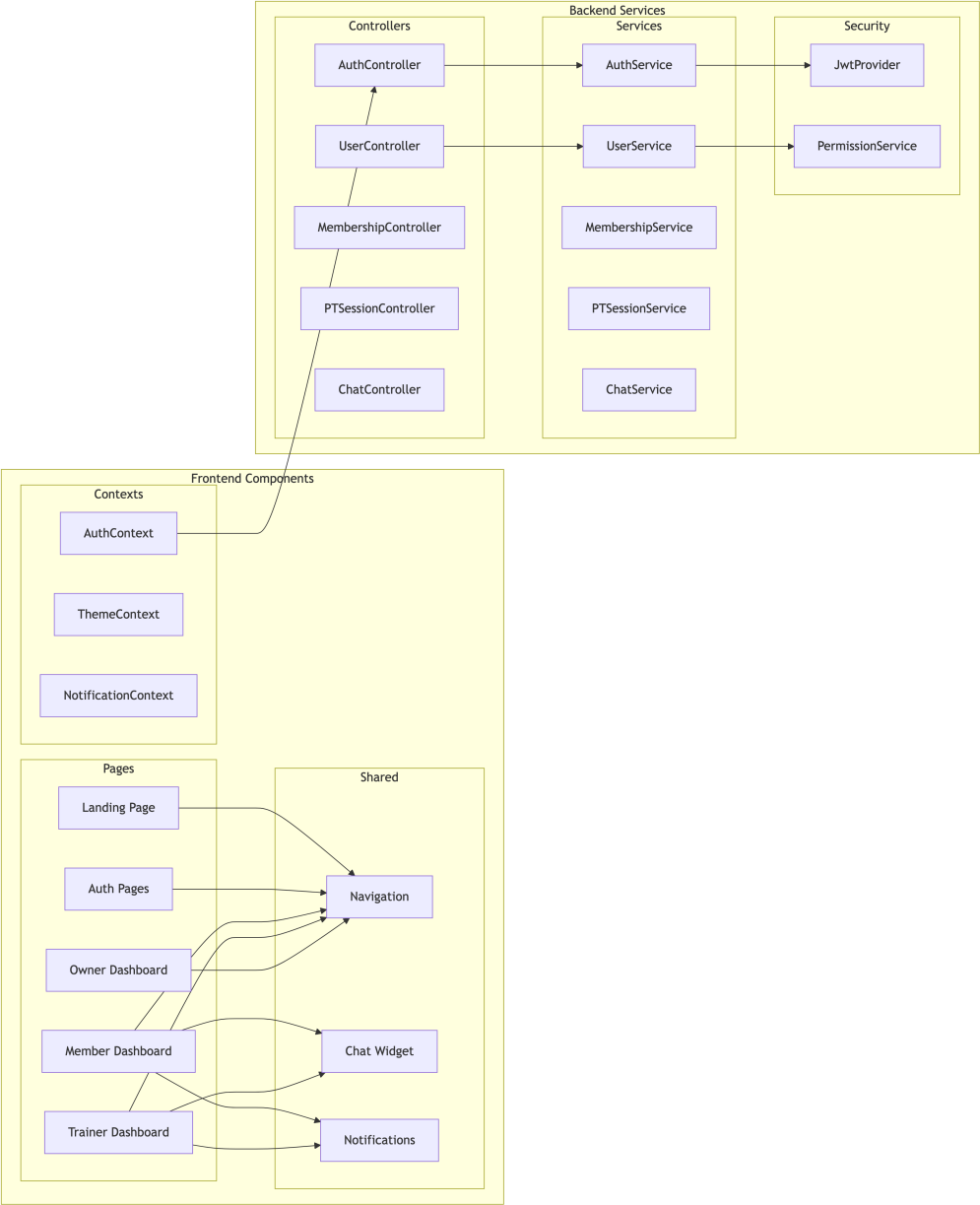
Technology	Purpose
Oracle Database	Primary Data Store
Spring Cache	Response Caching
HikariCP	Connection Pooling

External Services

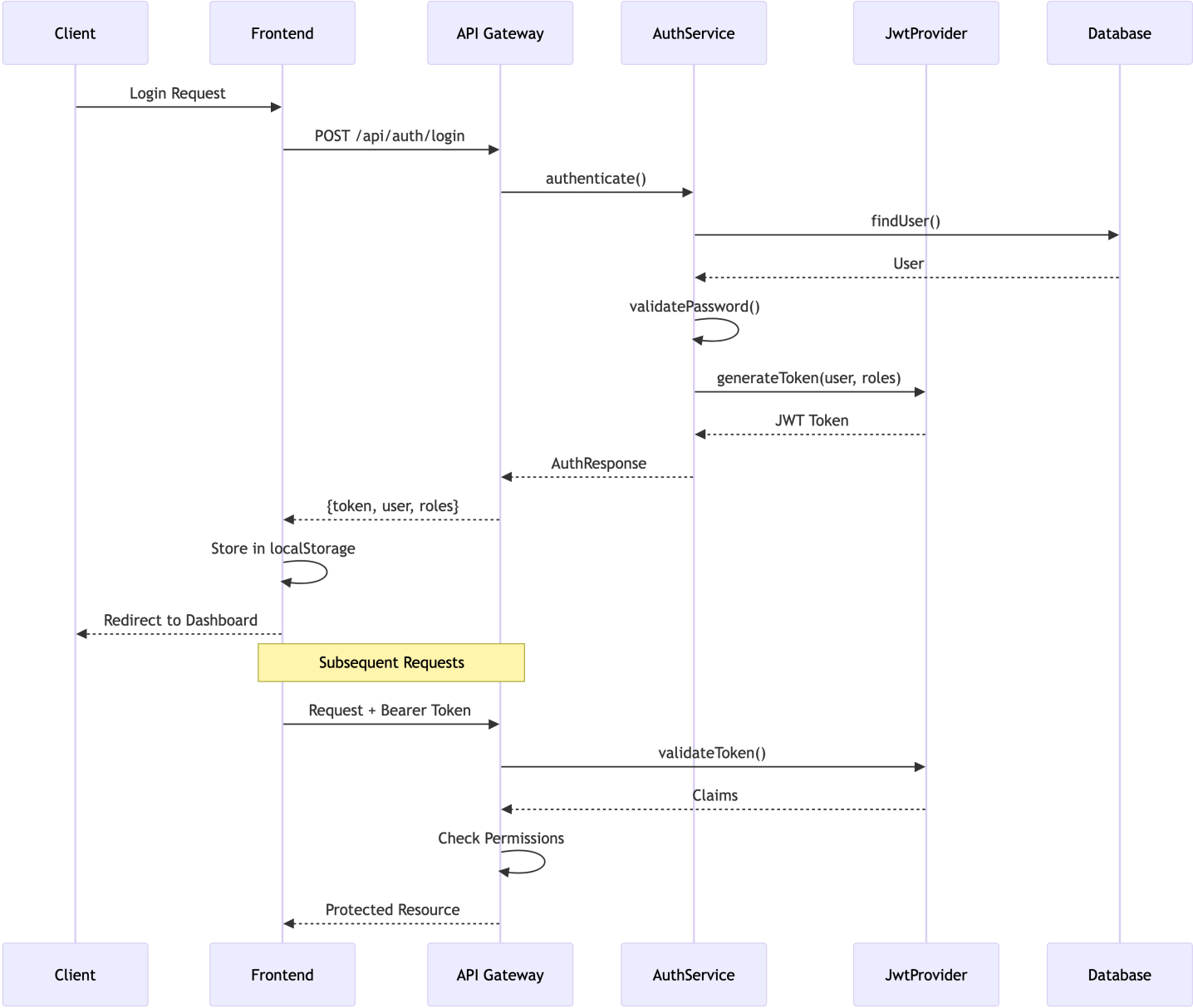
Service	Purpose
Google OAuth	Social Login

Service	Purpose
Facebook OAuth	Social Login
SMTP	Email Notifications
Twilio	SMS OTP

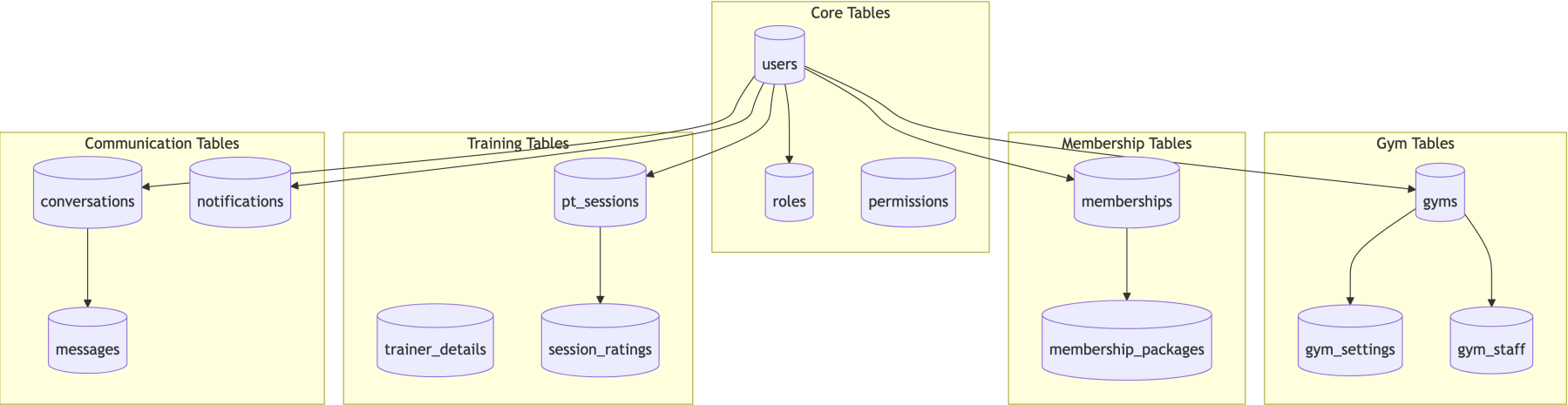
Component Architecture



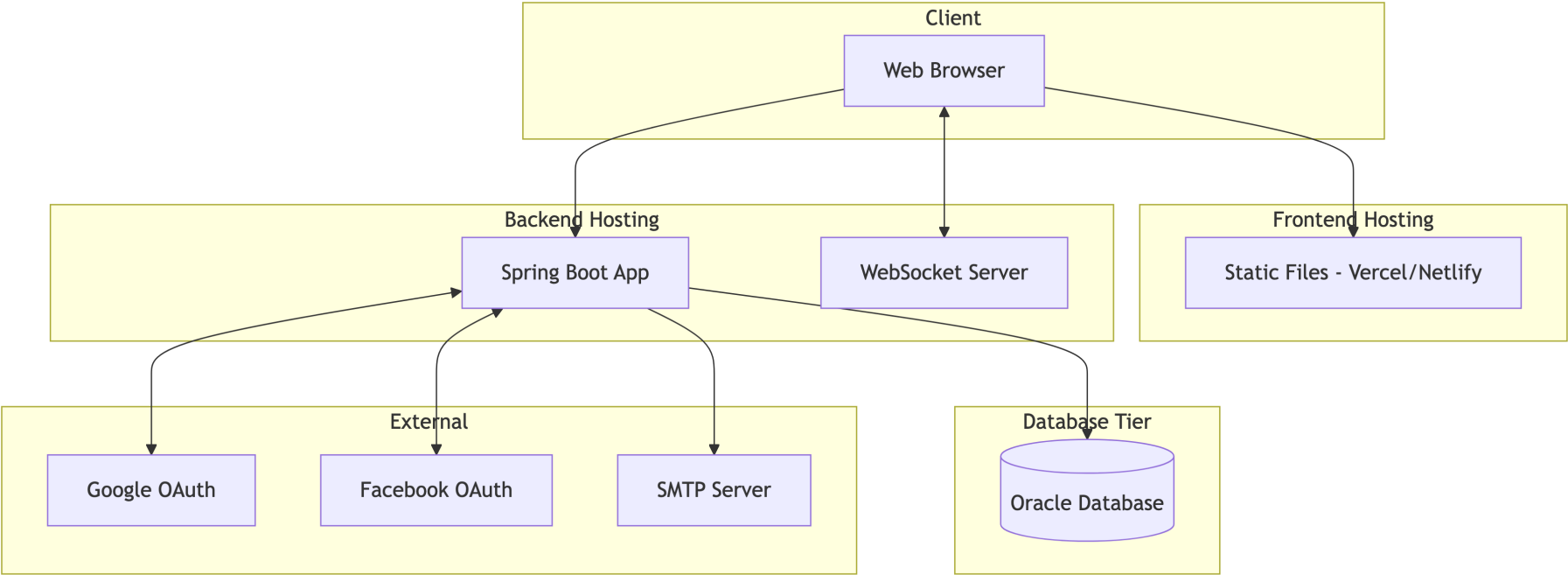
Authentication Flow



Database Architecture



Deployment Architecture



Security Architecture

Layer	Mechanism	Implementation
Transport	TLS/HTTPS	SSL Certificate
Authentication	JWT	Spring Security + JJWT
Authorization	RBAC	Custom Permission Service
Password	BCrypt	Spring Security
Session	Stateless	JWT Tokens
2FA	OTP	Email/SMS Verification
Data	Soft Deletes	Entity Annotations
CORS	Whitelist	Spring CORS Config

Scalability Considerations

1. **Multi-Tenancy:** Each gym is a separate tenant with isolated data
2. **Caching:** Spring Cache for frequently accessed data
3. **Connection Pooling:** HikariCP for database connections
4. **Lazy Loading:** JPA relationship optimization
5. **Stateless Auth:** JWT enables horizontal scaling
6. **WebSocket:** Dedicated connection management for real-time features